KD-DETR: Knowledge Distillation for Detection Transformer with Consistent Distillation Points Sampling

CVPR 2024

Yu Wang, Xin Li, Shengzhao Weng, Gang Zhang, Haixiao Yue, Haocheng Feng, Junyu Han, Errui Ding

- Problem/Objective
    - Knowledge Distillation in DETR-based Detector
- Contribution/Key Idea
    - Propose Distillation Points
    - Propose General-to-Specific Distillation Points Sampling Strategy

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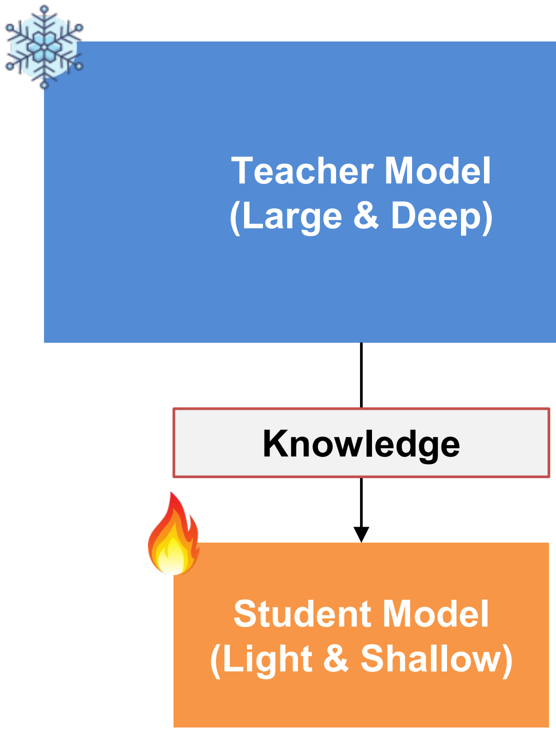
- **Knowledge Distillation**



잘 학습된 무거운 Teacher Model의 지식을 전달하여

가벼운 Student Model의 성능을 높이는 방법

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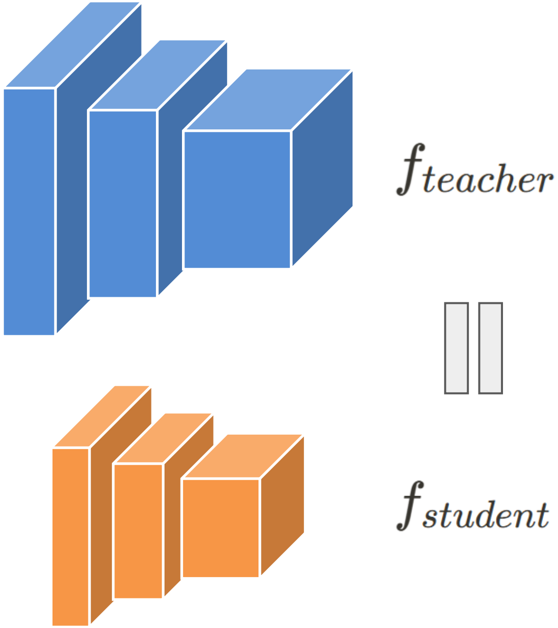
- Distillation Points

input $\mathbf{x}$



output $\mathbf{y}$

$$\mathbf{y} = f(\mathbf{x})$$

Need Sufficient & Consistent Distillation Points

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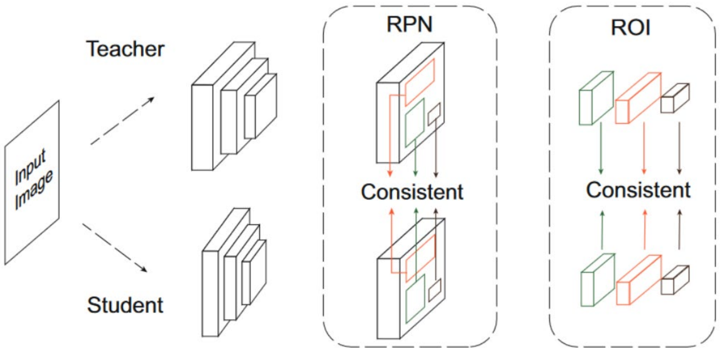
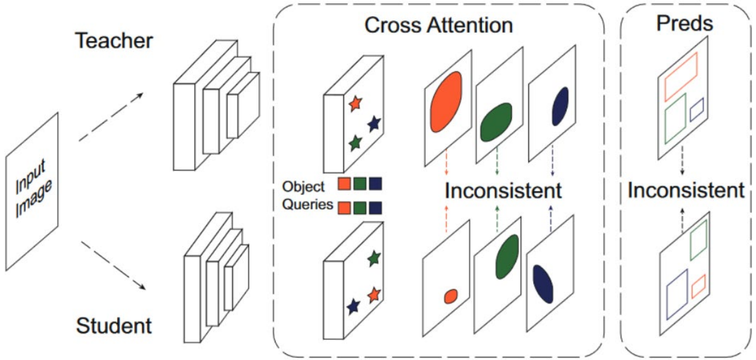
- **DETR Distillation Problem**



(a) Distillation Points in Classic Detector



(b) Distillation Points in DETR

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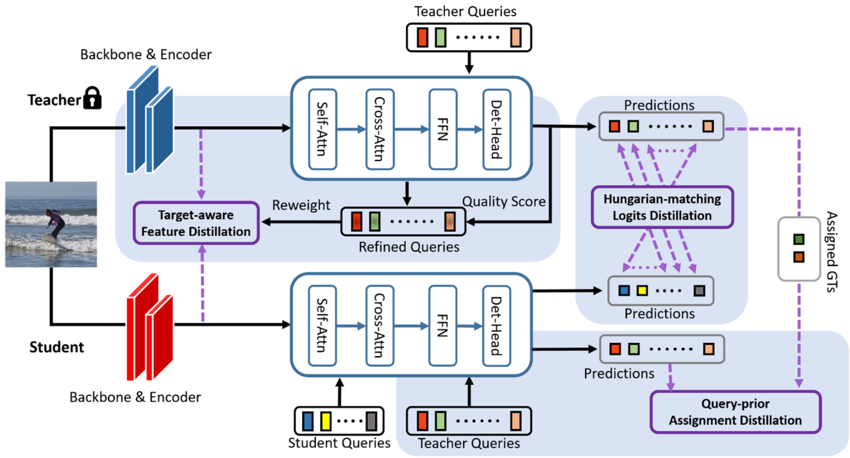
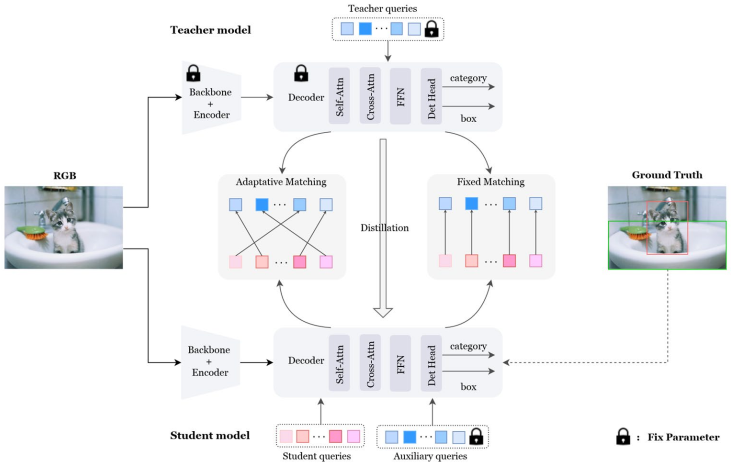
- Previous Methods Problems





[1] Chang, Jiahao, et al. "Detrdistill: A universal knowledge distillation framework for detr-families." *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023.
[2] Chen, Xiaokang, et al. "D $^ 3$ ETR: Decoder Distillation for Detection Transformer." *arXiv preprint arXiv:2211.09768* (2022).

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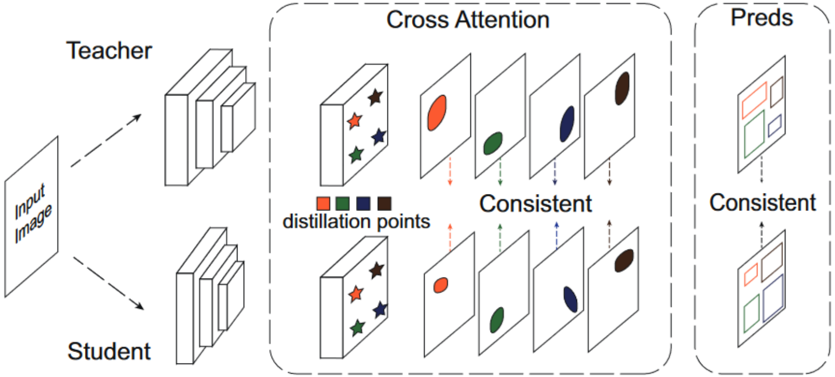
- Motivation



(c) Distillation Points in KD-DETR

How to construct
Sufficient & Consistent
distillation points in DETR?

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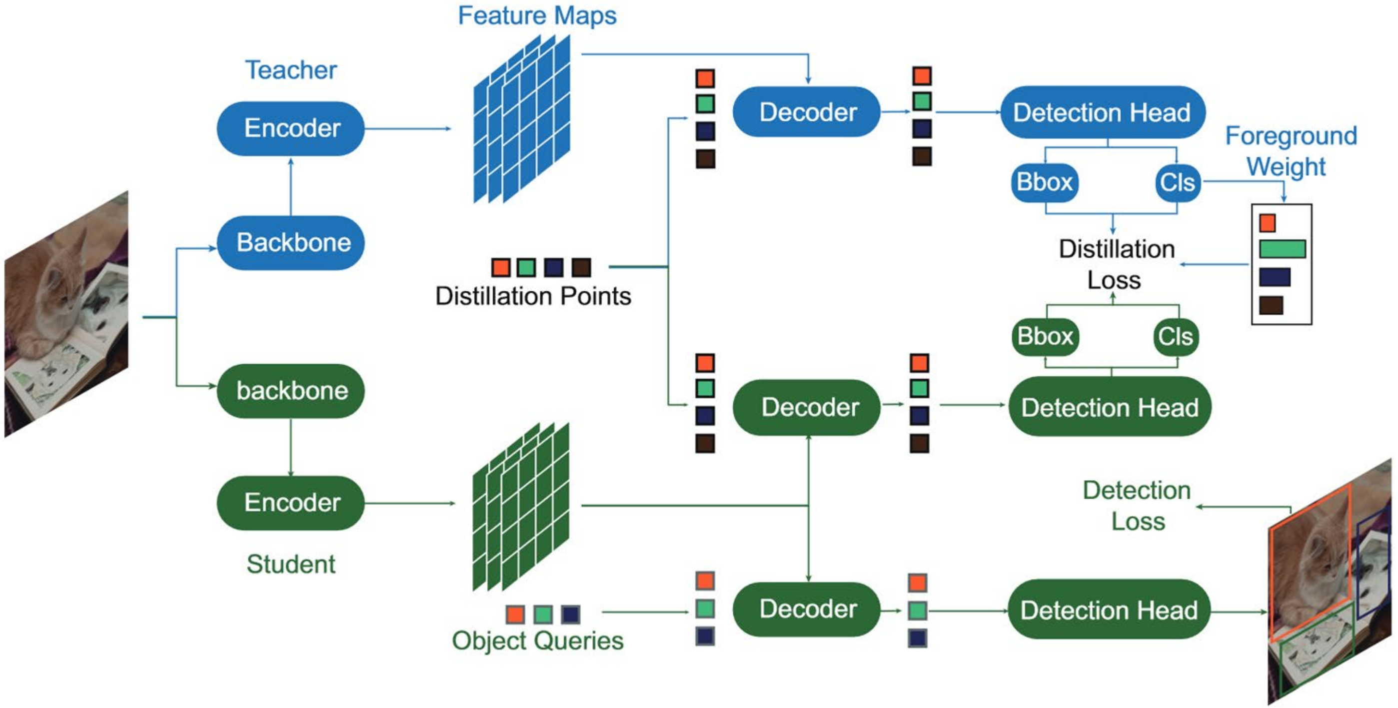
- **Overview**



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- Method

original input : $\mathbf{x} = \{\mathbf{I}, \mathbf{q}\}$

sampled distillation points: $\tilde{\mathbf{x}} = \{\mathbf{I}, \tilde{\mathbf{q}}\}$

Detection Task

$$\mathcal{L}_{det}$$

Distillation Task

$$\mathbf{c}^{\mathbf{s}}, \mathbf{b}^{\mathbf{s}} = f^s(\mathbf{I}, \tilde{\mathbf{q}})$$
$$\mathbf{c}^{\mathbf{t}}, \mathbf{b}^{\mathbf{t}} = f^t(\mathbf{I}, \tilde{\mathbf{q}})$$

$$\mathcal{L}_{distill} = \sum_{i=1}^{M}[\lambda_{cls}\mathcal{L}_{KL}(\hat{\mathbf{c}}_i^{\mathbf{t}}||\hat{\mathbf{c}}_i^{\mathbf{s}}) + \lambda_{L1}\mathcal{L}_{L1}(\mathbf{b}_i^{\mathbf{s}}, \mathbf{b}_i^{\mathbf{t}})$$

Total Loss

$$\mathcal{L} = \mathcal{L}_{distill} + \mathcal{L}_{det}$$

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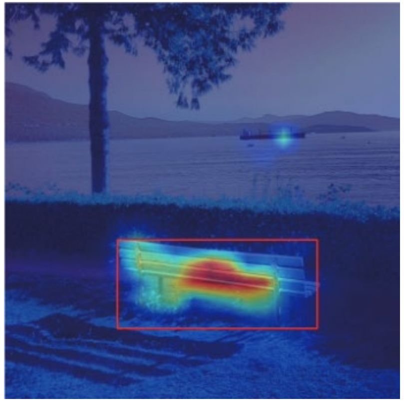
- Distillation Points Sampling

General Sampling with Random Initialized Queries

$$\mathbf{q_g} = \{\mathbf{q_i} \sim \mathcal{U}(0,1) | i = 1, \ldots, M_g\}$$

Specific Sampling with Teacher Queries

$$\mathbf{q_s} = \mathbf{q_{teacher}}$$



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- **Foreground Rebalance Weight**

이미지에서 객체 부분은 상대적으로 적고, 나머지는 대부분 배경
Knowledge Distillation 과정에서, Student Model이 배경에 치우쳐진 학습을 할 수 있음.
이러한 문제를 해결하기 위해 객체 부분에 더 많은 가중치를 부여

$$w_i = \max_{c \in [0,K]} p^t(y_c|\mathbf{q}_i)$$

$p^t(y_c|\mathbf{q}_i)$: Teacher model이 distillation point $\mathbf{q}_i$를 class c에 속할 것으로 예측한 확률

w_i : distillation point $\mathbf{q}_i$의 재조정된 가중치

K : 전체 class의 갯수

$$\mathcal{L}_{distill} = \sum_{i=1}^{M} \boxed{w_i} [\lambda_{cls}\mathcal{L}_{KL}(\hat{\mathbf{c}}_i^{\mathbf{t}}||\hat{\mathbf{c}}_i^{\mathbf{s}}) + \lambda_{L1}\mathcal{L}_{L1}(\mathbf{b}_i^{\mathbf{s}}, \mathbf{b}_i^{\mathbf{t}}) + \lambda_{GIoU}\mathcal{L}_{GIoU}(\mathbf{b}_i^{\mathbf{s}}, \mathbf{b}_i^{\mathbf{t}})]$$

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- **Experiments**

Strategy	AP	AP_{50}	AP_{75}
Baseline	36.2	56.1	37.9
Inconsistent	35.1	55.2	36.7
Similar Foreground	37.2	57.4	39.9
Similar General	37.4	58.0	40.6

Table 1. Distillation with Different Distillation Points. Inconsistent distillation points are harmful for distillation.

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- **Experiments**

Models		Epochs	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	GFlops	Params
DAB-DETR	ResNet-50(T)	50	42.1	63.1	44.7	21.5	45.7	69.3	90	44M
	ResNet-18(S)	50	36.2	56.1	37.9	16.9	39.0	53.5	49	23M
	Ours	50	41.4	61.4	44.2	21.2	44.7	58.7	49	23M
	Gains		**+5.2**	**+5.3**	**+6.3**	**+4.3**	**+5.7**	**+5.2**		
DAB-DETR	ResNet-101(T)	50	43.5	63.9	44.6	23.6	47.3	61.5	157	63M
	ResNet-50(S)	50	42.1	63.1	44.7	21.5	45.7	60.3	90	44M
	Ours	50	45.7	66.3	49.4	26.4	49.8	62.7	90	44M
	Gains		**+3.6**	**+3.2**	**+4.7**	**+4.9**	**+4.1**	**+2.4**		
Deformable-DETR	ResNet-50(T)	50	44.5	63.6	48.7	27.1	47.6	59.6	171	40M
	ResNet-18(S)	50	40.1	58.1	43.7	22.4	42.8	54.2	127	24M
	Ours	50	43.7	62.1	47.7	25.9	46.8	57.6	127	24M
	Gains		**+3.6**	**+4.0**	**+4.0**	**+3.5**	**+4.0**	**+3.4**		
Deformable-DETR	ResNet-101(T)	50	48.0	66.7	52.6	30.5	52.3	62.3	238	59M
	ResNet-50(S)	50	44.5	63.6	48.7	27.1	47.6	59.6	171	40M
	Ours	50	48.3	66.7	52.9	30.8	52.1	62.5	171	40M
	Gains		**+3.8**	**+3.1**	**+4.2**	**+3.7**	**+4.5**	**+2.9**		
DINO	ResNet-50(T)	36	50.9	69.0	55.3	34.6	54.1	64.6	245	47M
	ResNet-18(S)	12	44.0	61.2	48.1	27.4	46.9	56.9	200	30M
	Ours	12	48.4	65.5	53.0	31.6	51.7	62.3	200	30M
	Gains		**+4.4**	**+4.3**	**+4.9**	**+4.2**	**+4.8**	**+5.4**		
DINO	ResNet-101(T)	36	51.3	69.5	55.8	34.8	54.8	65.8	311	66M
	ResNet-50(S)	12	49.0	66.6	53.5	32.0	52.3	63.0	245	47M
	Ours	12	51.6	69.6	56.6	34.2	54.8	66.9	245	47M
	Gains		**+2.6**	**+3.0**	**+3.1**	**+2.2**	**+2.5**	**+3.9**		

Table 2. Results of the proposed KD-DETR with different DETR detectors and backbones with various scale.

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- **Experiments**

Method	AP	AP_S	AP_M	AP_L
Deformable DETR Res50	44.5	27.1	47.6	59.6
FGD[34]	44.1	25.9	47.7	58.8
FitNet[25]	44.9	27.2	48.4	59.6
DETRDistill[3]	46.6	28.5	50.0	60.4
Ours	**48.3**	**30.8**	**52.1**	**62.5**

Table 3. Comparison with state-of-the-art on Deformable DETR.

Method	AP	AP_S	AP_M	AP_L
RCNN-Res50	38.4	21.5	42.1	50.3
FGFI[29]	39.3	22.5	42.3	52.2
GID[6]	40.2	22.7	44.0	53.2
FGD[34]	40.4	**22.8**	44.5	53.5
Ours	**40.5**	22.7	**44.6**	**53.8**

Table 4. **Heterogeneous Distillation**: Distillation between DINO and Faster RCNN

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- Ablation Study

Object Queries	Distillation Points	AP	AP_{50}	AP_{75}
300	-	36.2	56.1	37.9
300+300	-	37.7	58.0	41.1
300	300	40.2	60.7	42.8

Table 5. **Benefit of Knowledge Distillation**: Simply increasing the set of object queries leads to trivial improvement

General	Specific	FRW	AP	AP_{50}	AP_{75}
			36.2	56.1	37.9
✓			38.7	59.1	40.7
✓		✓	39.9	60.2	42.5
	✓		40.0	60.5	42.4
	✓	✓	40.2	60.7	42.8
✓	✓	✓	41.4	61.4	44.2

Table 6. Distillation with Different Distillation Points Sampling Strategies. FRW refers to Foreground Rebalance Weight.

Number	10	50	100	300	900
AP	38.6	39.3	39.5	39.9	39.5
AP_{50}	58.3	59.2	59.3	60.2	60.4
AP_{75}	41.0	41.3	41.8	42.5	41.7

Table 7. Distillation with Different Number of General Distillation Points

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- Visualization of Distillation Points

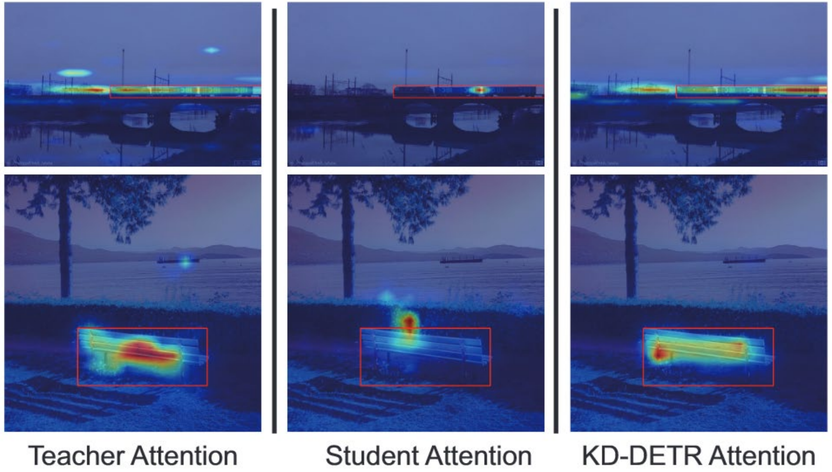

Figure 3. Attention Map Visualization of Specific Distillation Points: Images from left to right are from teacher, original student and student with KD-DETR. The corresponding predicted bounding box of the distillation points are marked with red rectangles.

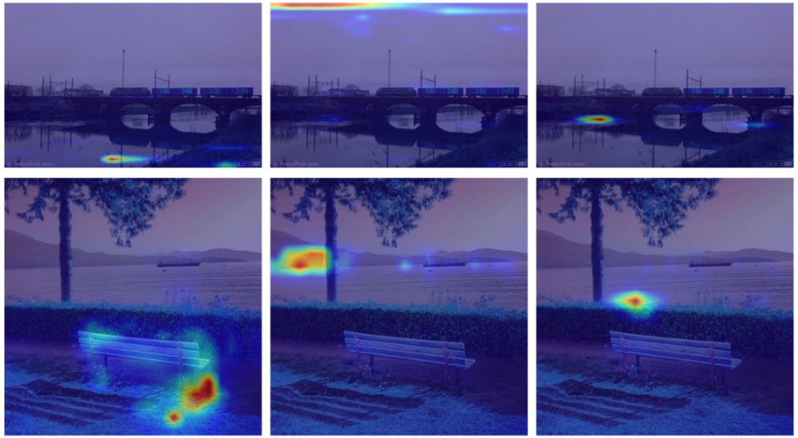

Figure 4. Attention Map Visualization of General Distillation Points: Teacher model's attention of general distillation points focuses more on the background regions, providing more information for student to mimic.